DeltaForge

An autonomous options agent on a validated momentum signal — slightly-ITM calls on the 30-minute cross, closed by the underlying's own levels.

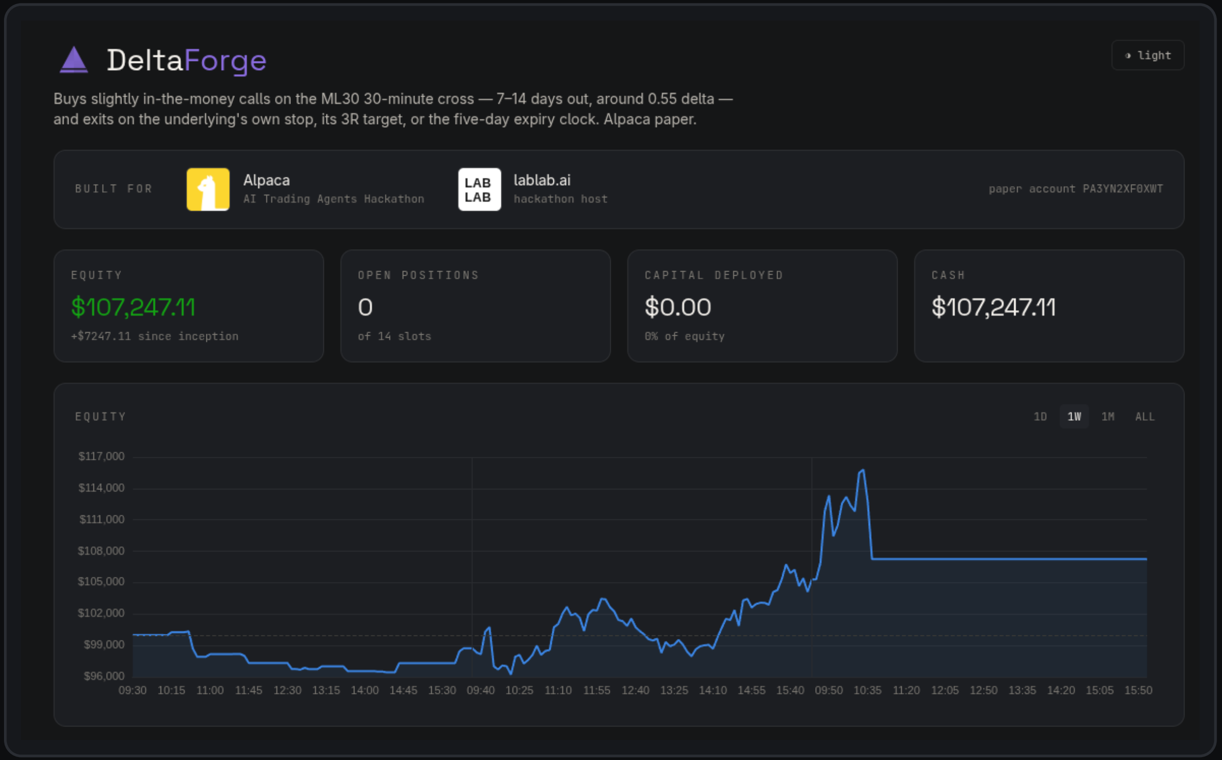
+7.25%	**2,345**	**13%**
4 SESSIONS · PAPER $100K	BACKTESTED TRADES	TARGETS HIT — THE EDGE IS THE STOP

Alpaca AI Trading Agents Hackathon · lablab.ai · deltaforge.geekendzone.net

THE RESULT

+7.25% in four sessions, closed to cash.

One autonomous bot, one brand-new Alpaca paper account (PA3YN2XF0XWT), $100,000 at inception on Aug 31. Every position was opened by the agent; the book was closed to cash on judging day by a coded protective floor. The number is frozen — cash cannot be re-marked.



$107,247
EQUITY AT CLOSE

+$7,247
SINCE INCEPTION

0 / 14
POSITIONS OPEN · SLOTS

$336
FRICTION TO CLOSE, 6 LEGS

HOW IT TRADES

A validated 30-minute cross hands the agent three anchors.

The ML30 signal fires when a single 30-minute candle closes **above its 55-period SMA** having been below it the bar before, sits **above the 21-period SMA**, and the candle itself is bullish.

Anchor	Rule	Role
Direction	long, at the cross bar's close	what to buy
Stop	lowest low of the 8 bars before entry — frozen	when the thesis is wrong
Target	entry + 3× the stop distance	how far "right" is worth riding

Instrument: a slightly-ITM call, 0.55 delta, 7–14 DTE, $7,000 a position across 14 slots. The underlying's own levels close the trade — stop touch, target touch, or a DTE clock at 5 days to expiry. Signals whose 3R target sits under 5% away are refused: a short walk cannot pay for an option's bid-ask and theta.

THE ANCHORS, AS THE DASHBOARD DRAWS THEM

NVDA — the cross, the frozen stop, the 3R target.

The 21 (blue) and 55 (orange) SMAs are the ones the bot computes — not a reconstruction. The ENTRY candle is the cross; the red line is the 8-bar pivot stop; the green line is 3R. Below: breakeven, asked vs filled, delta at entry, max loss.

WHERE THE EDGE COMES FROM

Not from hitting targets. From what a stop costs a call.

Measured over **2,345 backtested trades** at the live configuration, on real Alpaca options data, Feb 2024 ⟶ Aug 2026:

52%	13%	34%
STOP OUT · −36.8% OF DEBIT	HIT THE 3R TARGET · +119.7% OF DEBIT	EXIT ON THE CLOCK · WHAT THE MOVE GAVE

Expectancy: +9.9% of debit per trade

A long call cannot lose more than was paid for it, and still carries time value when the underlying touches the stop. Truncating the losses that way — while leaving the right tail open — is the entire trade. Over the full window the same rules turn **$3,000 into $12,130 (+304%)** against **+66%** for the underlying shares. The per-trade edge is identical at every position size tested (`reports/budget_100k_v2/`); what sizing buys, and costs, is concentration.

WHAT DID NOT SURVIVE

The first thesis was killed by its own verification.

The repo's original structure was a **bull call debit spread** with its short strike at the 3R target — the literal translation of the system into options. It backtested at **+286%**.

Verification killed it. Its profit rested on **5 trades out of 772** and on a cohort of implausibly cheap spreads that contributed more than all of the gain. The long call replaced it, and the post-mortem stayed public.

> ***"A backtest you cannot falsify is marketing."***
>
> docs/BACKTEST.md · What did not survive verification

+286%

THE DEBIT SPREAD'S HEADLINE RETURN

5 / 772

TRADES CARRYING THE PROFIT

144

CONFIGURATIONS SWEPT BEFORE TRUSTING IT

THE AGENT

The live executor mirrors the backtest bar for bar.

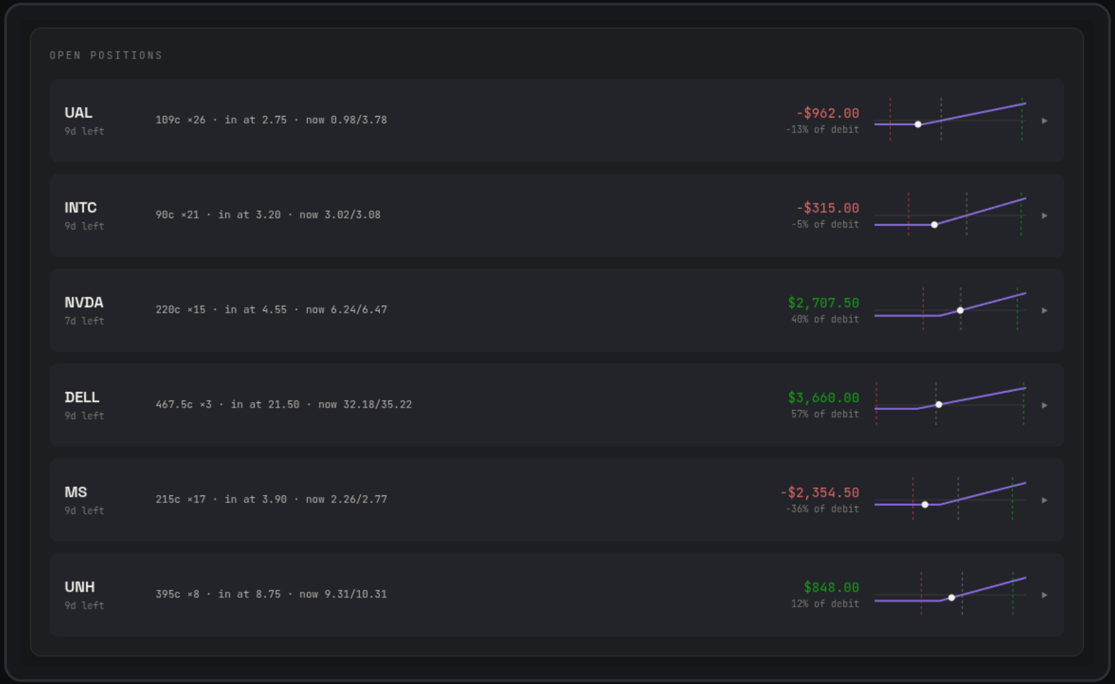
- **Same entry logic** imported from the validated signal code, same frozen stop, same-bar precedence, same DTE clock — any divergence would make the paper phase measure something other than what was tested.
- What the backtest could not model, the bot **measures**: limit orders at mid, chased no further than the haircut the backtest assumed, **abandoned rather than crossed**.
- Every decision — signal, refusal, order, fill, exit — journals to **SQLite** and renders on a **public Next.js dashboard**: the bot's own SMAs and levels, the entry candle, and a history that includes the trades that never filled.
- The **Alpaca Trading API** for orders, **Market Data** for option chains and SIP/IEX bars; one Python process under systemd on a self-hosted homelab, published through a Cloudflare Tunnel.

What the journal keeps for every position: contract, delta at entry, asked vs filled on both legs, fees, the three levels, and the fill log.

THE JUDGED WEEK, HONESTLY

Why one week undersells it — and what it did show.

- **The window is shorter than one trade's life.** Median hold is 3.8 days; exits arrive on a 5–9 day clock.
- **The book needs over a week to reach steady state.** It fills at ~4 positions a day from flat; judging caught it mid-ramp.
- **The edge lives in the right tail.** At a 36.8% win rate, expectancy emerges over hundreds of trades — not eight.
- **End-of-day marks understate a long-options book.** Alpaca marks long options at the bid: a $5,165 gap to mids at the Sep 2 close.
- **The floor lesson is on the record.** The judging-day floor was calibrated tighter than the book's own measured volatility and fired at the bottom of a V. That study is first in the queue.



The book mid-ramp on Sep 2: six of fourteen slots filled, winners and losers side by side.

EVERYTHING IS PUBLIC

The repo, the backtests, the journal, the dashboard.

LIVE DASHBOARD	https://deltaforge.geekendzone.net
REPOSITORY	https://github.com/jacedeno/deltaforge
BACKTEST REPORT	docs/BACKTEST.md · every artifact in reports/, reproducible from the CLI
STRATEGY ANALYSIS	docs/ANALYSIS.md · why a long call, and what was rejected
OPERATIONS RECORD	docs/DEPLOYMENT.md · the judged session, the floor, the honest notes
ALPACA PAPER ACCOUNT	PA3YN2XF0XWT · opened for the event, $100,000 inception 2026-08-31
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Built in six days inside the hackathon window (first commit 2026-08-28 14:41 CDT). The ML30 signal — the 21/55 cross, the 8-bar pivot stop, the 3R target — is the author's prior research and is disclosed as such; the overlay analysis, the backtest engine, the paper bot and the dashboard were all built for this event. Paper trading, simulated funds, real market data. Nothing here is investment advice.